



1. Find an isomorphism $f : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$ that is different from the identity map.
2. Let $\mathbb{R}^\times := \mathbb{R} \setminus \{0\}$. Find all linear polynomials $P(x) := a_1x + a_0$ such that the map $f_P : (\mathbb{R}^\times, \cdot) \rightarrow (\mathbb{R}^\times, \cdot)$ defined by $f_P(\alpha) = P(\alpha)$ is a homomorphism. Do the same exercise for quadratic polynomials $P(x) := a_2x^2 + a_1x + a_0$. What about higher degree polynomials?
3. Consider $\text{rot}(P_5)$, the group of rotations of a regular pentagon. Suppose G is a group and $f : \text{rot}(P_5) \rightarrow G$ is a homomorphism. Argue that either $f(x) = 1_G$ for every $x \in \text{rot}(P_5)$, or f is an injective map.
4. Let G be a group generated by two elements $a, b \in G$ with the following relations: $a^2 = 1, b^2 = 1, (ab)^2 = 1$. Argue that there is an isomorphism from G to the group of symmetries of a rectangle.
5. Consider the binary operation $*$ on $G = \{1, a_1, a_2, a_3\}$ given by the following:

$*$	1	a_1	a_2	a_3
1	1	a_1	a_2	a_3
a_1	a_1	1	a_3	a_2
a_2	a_2	a_3	1	a_1
a_3	a_3	a_2	a_1	1

Show that G is a group under the operation $*$. Consider the set

$$X := \{(x_1, x_2, x_3) : \text{each } x_i = 1 \text{ or } -1\}.$$

Show that $\cdot : G \times X \rightarrow X$ defined by

$$\begin{aligned} 1 \cdot (x_1, x_2, x_3) &= (x_1, x_2, x_3) \\ a_1 \cdot (x_1, x_2, x_3) &= (x_1, -x_2, -x_3) \\ a_2 \cdot (x_1, x_2, x_3) &= (-x_1, x_2, -x_3) \\ a_3 \cdot (x_1, x_2, x_3) &= (-x_1, -x_2, x_3) \end{aligned}$$

is a group action. Find the orbit of $(1, 1, 1)$ in this action. How many orbits are there?

6. Consider A_4 , the group of even permutations on $\{1, 2, 3, 4\}$. Count the number of orbits of the action

$$A_4 \times \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$$

given by $(\sigma, i) \mapsto \sigma(i)$.

7. An action $G \times X \rightarrow X$ is called *transitive* if it has only one orbit. Give an example of transitive action, and an example of non-transitive action.